CH 1 — Introduction to Probability

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1.1 Definition of Probability

1.2 Mathematical Probability Model

📘 **Definition — Baisc Definitions**

An **experiment** is a repeated process with multiple **trials** where different **outcomes** can occur.

A **sample space** S is a **set** of all possible outcomes on one tiral of an experiment can be

- Discrete($\mathbb{Z}$, countable)
- Continuous($\mathbb{R}$, $\mathbb{C}$, uncountable)

An **event** is a subset of the **sample space**(i.e $A \subseteq S$, it can be $\varnothing$, S, or anyother) and can be

- Simple event(contain 1 element)
- Compound event(contain > 1 element)

The probability of a simple test a_i is $P(a_i) \in \mathbb{R}$, s.t.

- $0 \leq P(a_i) \leq 1$
- $\sum_{a_i \in S} P(a_i) = 1$

We say the $\{P(a_i), i = 1, 2, ...\}$ is the **probability distribution** on S

If S is finite and has equally likely outcomes, then $P(a_i)$

The probability of an event A is $P(A) = \sum_{a_i \in A} P(a_i)$, the sum of the probabilities of the simple event that make up A

If S is finite with equally likely outcomes, then $P(A) = \frac{|A|}{|S|}$ (consistent with classical definition)

The odds in favour of an event A are defined as $\frac{P(A)}{1-P(A)}$

The odds against in favour of an event are defined as $\frac{1-P(A)}{P(A)}$

1.3 Counting Rules

Info — Counting Rules

Addition rule: if we can do job 1 in p ways and job 2 in q ways, then we can do job 1 **or** 2 in $\boldsymbol{p+q}$ ways.

Multiplication rule: if we can do job 1 in p ways and job 2 in q ways, then we can do job 1 **and** 2 is $\boldsymbol{pq}$ ways

Sampling WITH replacement: can select the same object multiple times.

- The sample space size is n^k

Sample WITHOUT replacement: cannot select the same object multiple times.

Permutation: sample without replacement **with order**

- The sample space size is $n^{(k)}$: $(n)(n-1)...(n-k+1)$ or $\frac{n!}{(n-k)!}$

Combination: ways of selecting k objects from n without replacement, and order does not matter

- $\frac{n^{(k)}}{k!} = \binom{n}{k} = {}_nC_k$

Example:

1. Of the 36 outomes when two 6-sided dice are rolled, how many ways are there to have equal values on both dices or a total of 5?

   ANS: 10

2. In Texas poker. Say you already have 6, 7, 9, 10. What of the probability of pulling 8? If not on turn, what about next two to obtain 8?

   ANS: $\frac{4}{47}, \frac{4}{46}, \frac{4}{45}, \left(\frac{4}{47} \times \frac{43}{46} + \frac{43}{47} \times \frac{4}{46} + \frac{4}{47} \times \frac{3}{46}\right)$

3. IP address is 4 numbers between 0-255. How many potential IP exsits? Probability at least one is even? Probability of all numbers are different

   ANS: $256^4, 1 - \frac{1}{2^4}, 246^{(4)} = \frac{255}{256} \times \frac{254}{256} \times \frac{253}{256}$

4. Lotto 6/49. Each ticket has 6 numbers choosen from 1-49.

   - How many possible tickets are there?

     No repetition where order does not matter: $\binom{49}{6} = 13983816$

   - Number of tickets that match 5 of 6 numnbers

     5/6 are winning numbers, and 1 number should be 1 of 43 non-winning numbers: $\binom{6}{5}\binom{43}{1}$

5. A lock as a 4-digit code

   - How many have all different numbers?

     $10^{(4)} = 10 \cdot 9 \cdot 8 \cdot 7 = 5040$

   - How many odd numbers over 9000?

     $1 \cdot 10 \cdot 10 \cdot 5 = 500$

- Both at the same time?

  $1 \cdot 8 \cdot 7 \cdot 4 = 224$

- Strictly decreasing order?

  $\binom{10}{4} \cdot 1 = 210$

- How many include 3 different digits?

  $10 \cdot 9 \cdot 8 \cdot \binom{4}{2} = 4320$, 10 choices for pairs, $9 \cdot 8$ for remaining digits choice. There are $\binom{4}{2}$ ways of positioning those pairs as they do not have to be consective

  or $\binom{10}{3} \times \binom{3}{1} \times \binom{4}{2} \times 2$, pick 3 out of 10 numbers, 3 different numbers, $\binom{4}{2}$ ways of positioning pairs, 2 ways of ordering the singles

  or $10 \times \binom{9}{2} \times \frac{4!}{2!1!1!}$

📘 **Definition — Pascal Triangle**:

- $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$
- Symmetrical, $\binom{n}{k} = \binom{n}{n-k}$
- Start and end with 1, $\binom{n}{0} = \binom{n}{n} = 1$
- Rows' sum is in form 2^n, $\sum_{k=0}^{n} \binom{n}{k} = 2^k$
- n-th row is the coefficient of each term of expanded form $(a+b)^n$

Ways of positioning "STATISTICS" We have $\binom{10}{3}$ ways of putting "S"

We have then $\binom{7}{3}$ ways of putting "T"

We have $\binom{4}{2}$ ways of putting "I"

We have $\binom{2}{1}$ ways of putting "A"

We have $\binom{1}{1}$ way of putting "C"

$\binom{10}{3} \times \binom{7}{3} \times \binom{4}{2} \times \binom{2}{1} = \frac{10!}{3!7!} \times \frac{7!}{3!4!} \times \frac{4!}{2!2!} \times \frac{2!}{1!1!} \times \frac{1!}{1!0!} = \frac{10!}{3!3!2!1!1!}$

📘 **Definition — Multinomial coefficient**

If we arrange n objects of k types where there are n_1 of type 1, ..., n_k of type k. There are

$$\frac{n!}{n_1!...n_{k!}}$$

ways of result could look like. This quantity is the multinomial coefficient.